

Data Science Project

CHEMICAL CONTENT BASED RECOMMENDATION SYSTEM - SEPHORA



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Problem Statement

- Difficulty in selecting a beauty product or personal care product due to its complex structure of ingredients.
 - Lack of expertise of customers to interpret the chemical composition which are suitable for their skin type especially people with sensitive skin.
 - Current selection of products by consumers involves trial and error method, which might prove harmful for people with sensitive skin.
 - No easier or data driven way to compare products based on their ingredients.
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Project Overview

Objective:

- Creating a model that would recommend product using their chemical composition (ingredients) by comparing them.

Approach:

- Standardizing the list of ingredients
- Building a matrix of these ingredients
- Measuring the similarity between these ingredients

Method Used:

- Tokenization of ingredients and then creating Document-Term Matrix.
- Using t-SNE for dimensionality reduction
- Using Bokeh for interactive visualisation

Result:

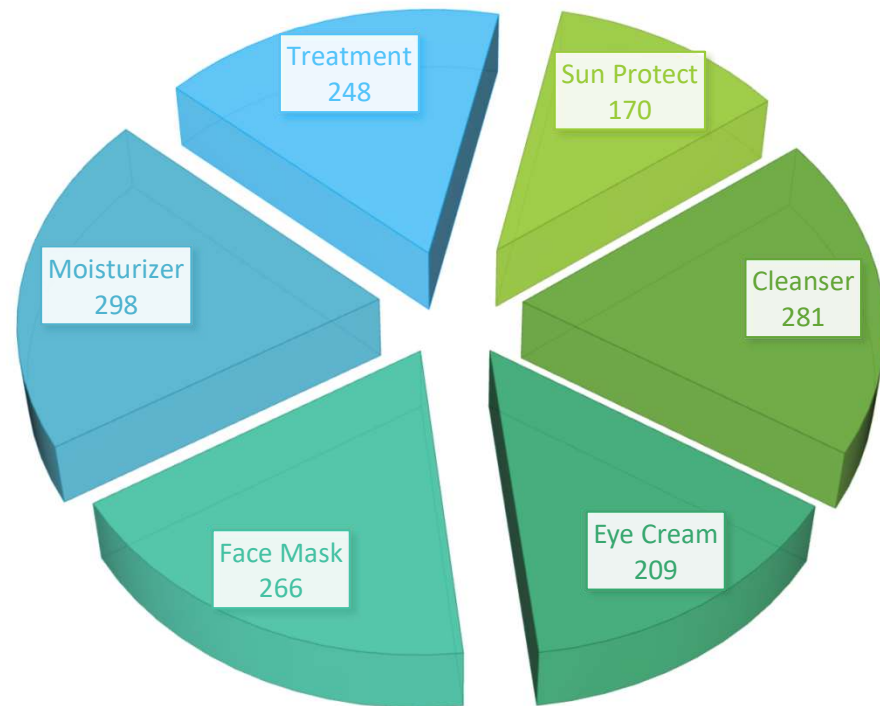
- Scatter plot for mapping product similarity
- Similar ingredients profiles on scatter plot being closer to each other.
- Help consumers to compare moisturizer for dry skin.
- Support an informed selection of the product

Dataset Overview

Dataset:

- Rows: 1472
- Columns: 11
- Product categories: 6
- Skin Type: 5

**PRODUCTS &
COUNT**



Data Processing

- Selecting dataset with the required skin (dry) type and product category (Moisturizer)
- Missing and Inconsistent data are removed.
- The text are standardized into lower cases using `.lower()`
- Ingredients we further categorized into individual components using “Tokenization”
- A vocabulary of all unique ingredients were mapped with an integer index (`ingredient_idx`) creating a dictionary.
- A foundation for feature engineering was created by storing the ingredient index and corpus of tokenized list.

Feature Engineering

- The list of unique ingredients were mapped to index Values(ingredient_idx).
- A corpus of tokenized ingredients list was created.
- Converted the list of ingredients into Binary vectors(One-hot Encoding).
- Document term matrix was created for all products.

```
: # Get the number of items and tokens
M = len(moisturizers_dry)
N = len(ingredient_idx)
print(M)
print(N)

# Initialize a matrix of zeros
A = np.zeros((M, N))
print(A)
```

190

2233

Modelling and Key Techniques

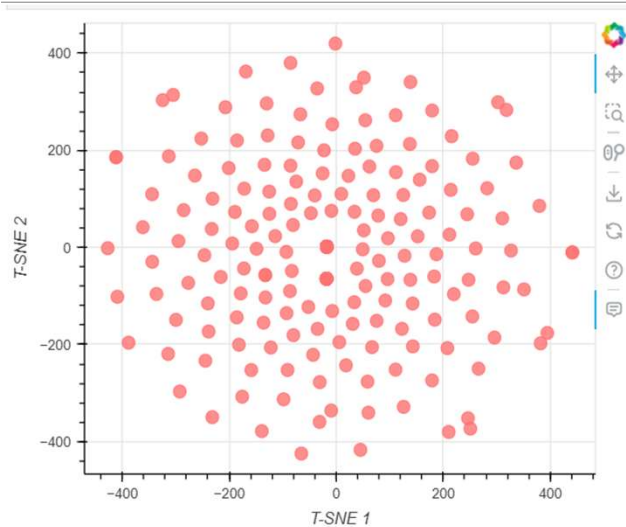
t-SNE

- t-SNE was used to reduce the dimensionality of high dimensional ingredient data.
- Products were projected in 2D space based on ingredient similarity.
- Similarity based approach was used to compare cosmetic products.
- For interactive exploration and visualization Bokeh was used.

Bokeh Visualisation

- Scatter plot is plotted
- X-axis= t-SNE 1, Y-axis= t-SNE 2
- Hover Tool: Brand, name, price, Item name
- Exploratory Comparison without code
- Plot id linked to filtered Dataframe

Visualisation

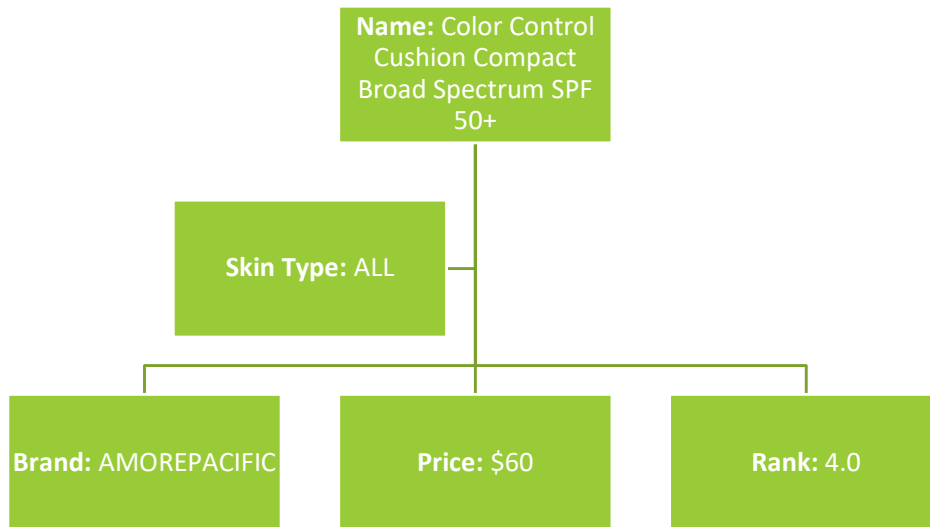


Observations:

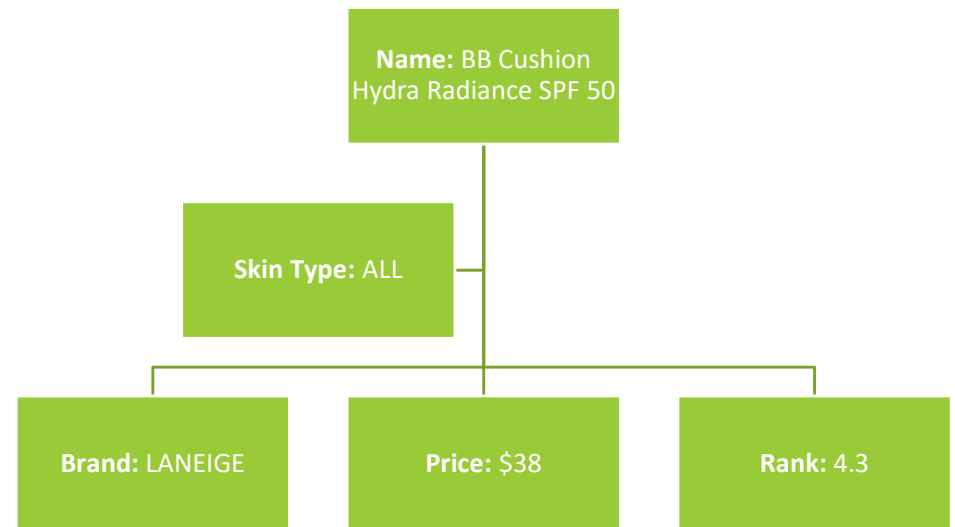
- The number of dry-skin plotted is 190 with colour “#FF7373”.
- Both axes project the range of +/- 400 expressing the ingredient diversity.
- It can be seen that there are no tight clusters,
- The cursor could be hovered over the dots to find out Item, Brand, Price, Rank.

Results and insights

PRODUCT WE LIKED



RECOMMENDED ALTERNATIVE



Limitations and Future improvements

LIMITATIONS

- Conducted analysis only on moisturizer for dry skin.
- T-SNE model does not explain as to why are similar. It only projects them similar because the model considers them neighbours in a high dimensional space.
- The concept of concentration levels, ingredients interaction and ordering are left out while using One-hot encoding as it treats the presence of ingredients as equal.

FUTURE IMPROVEMENTS

- We can add a database for skin safety ingredients database to potentially flag the ingredient.
- We can apply the same pipeline to filter across the variety of products and skin type for comparison eventually creating a universal cross category recommender.
- Can work on a new/live data pipeline to scrape newer data as it comes, keeping the recommendations current.

Conclusion

- Ingredients are converted in binary data using tokenisation and one hot encoding so that they could be computable.
- Using t-SNE to compress 2233 ingredients feature matrix into a 2-D scatter plot, preserving the product similarity amongst 190 items.
- The model concluded that “BB Cushion Hydra Radiance SPF 50” a product of “LANEIGE” identical to AMOREPACIFIC’s Color Control Cushion Compact Broad Spectrum SPF 50+ and its even cheaper to it. The moisturizer of AMOREPACIFIC was \$60 where as the moisturizer of LANEIGE was \$38 which was \$22 lesser.
- Without the knowledge of chemistry of different ingredients consumers can now choose moisturizer by spatial proximity alone.

	Label	Brand	Name	Price	Rank	Ingredients	Combination	Dry	Normal	Oily	Sensitive
45	Moisturizer	AMOREPACIFIC	Color Control Cushion Compact Broad Spectrum S...	60	4.0	Phyllostachis Bambusoides Juice, Cyclopentasil...	1	1	1	1	1
	Label	Brand	Name	Price	Rank	Ingredients	Combination	Dry	Normal	Oily	Sensitive
55	Moisturizer	LANEIGE	BB Cushion Hydra Radiance SPF 50	38	4.3	Water, Cyclopentasiloxane, Zinc Oxide (CI 7794...	1	1	1	1	1